A Survey and an Empirical Evaluation of Multi-View Clustering Approaches

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Multi-view clustering (MVC) holds a significant role in domains like machine learning, data mining, and pattern recognition. Despite the development of numerous new MVC approaches employing various techniques, there remains a gap in comprehensive studies evaluating the characteristics and performance of these approaches. This gap hinders the in-depth understanding and rational utilization of the recently developed MVC techniques. This study formalizes the basic concepts of MVC and analyzes their techniques. It then introduces a novel taxonomy for MVC approaches and presents the working mechanisms and characteristics of representative MVC approaches developed in recent years. Moreover, it summarizes representative datasets and performance metrics commonly employed for evaluating MVC approaches. Furthermore, we have meticulously chosen 35 representative MVC approaches to conduct an empirical evaluation across seven real-world benchmark datasets, offering valuable insights into the realm of MVC approaches.

CCS Concepts: • **General and reference** → *Document types*; *Surveys and overviews;* • **Computing methodologies** → *Machine learning*; *Learning paradigms*; *Unsupervised learning*; *Cluster analysis*;

Additional Key Words and Phrases: Multi-view clustering, consensus and complementary principles, information fusion, weighting, clustering routine

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1 INTRODUCTION

With advances in information acquisition technologies, multi-view data, collected from diverse domains or obtained from various feature extractors, has become ubiquitous. Each domain or feature is referred to as a particular view, which reflects a specific property of an object—for example, color and texture describe an image from two different aspects. Different views generally provide compatibility and complementary information to each other [1–4]. The effective utilization of this information can facilitate an increase in the accuracy of learning algorithms [5].

One way of solving the multi-view problem is to treat each view independently, or to concatenate vectors from different views into a new vector and then to apply single-view learning algorithms straightforwardly on the concatenated vector. The strategy that treats each view independently may overlook the complementary and diverse information of different views, thereby losing the advantages of multi-view data. However, the strategy of concatenation disregards the specific properties of each view and may result in high dimensionality, potentially reducing the interpretability of different views and causing overfitting on a small training sample [2, 6]. Unlike the aforementioned strategies, **Multi-View Learning (MVL)** seeks to integrate features and structural information from multiple views, exploiting the richer properties of data to enhance learning performance. Therefore, MVL has emerged as an important research area [4, 5, 7–10].

Multi-View Clustering (MVC), playing an imperative role in MVL for reducing data complexity and facilitating interpretation, aims to group samples (objects/instances/points) with similar feature structures or patterns into the same group (cluster) and samples with dissimilar ones into different groups, by combining the available feature information from different views and searching for consistent clusters across different views [11, 12]. As a powerful alternative learning tool for uncovering the underlying structure shared by multiple views in the absence of label information, MVC has attracted increasing attention in recent years and has been successfully used in widespread applications [13]. However, MVC is not a trivial task. On one hand, each view may have its own feature or distribution, and multiple views may take different forms as well as exhibit heterogeneous and diverse properties. Moreover, in some cases, the data in each separate view may not be compatible with the others [14]. These differences pose a challenge to integrate information from complex distribution and diversified heterogeneous features of multi-view data to obtain the true categories of the data. If a clustering approach is unable to cope appropriately with multiple views, these views may even degrade the performance of MVC. On the other hand, the clustering approach needs to elaborately design a way to maximize clustering quality within each view while simultaneously making clustering results across different views as consistent (agree with each other) as possible. When multiple sets of features are available for each individual sample, how to express the relationship of multiple views and how to integrate these views to identify essential grouping structure are two important questions needed to be answered by MVC [3]. In addition, the multi-view data collected in the real world are often incomplete (missing samples or features), uncertain, or dynamic. This is due to the complexity in data collection and transmission. The challenges introduced by missing samples or features, uncertainty, and dynamic changes add complexity to MVC. Therefore, **Incomplete Multi-View Clustering (IMVC)**, uncertain MVC, and dynamic MVC have also been developed.

To meet these challenges, many research efforts have been made and a number of MVC approaches have been developed, such as co-regularized multi-view spectral clustering [15],

Non-negative Matrix Factorization (NMF)-based MVC [16], and bipartite graph-based multi-view spectral clustering [17, 18]. These approaches employ different techniques to tackle the MVC issues from different viewpoints. In addition, several survey studies have been conducted to investigate the theories and techniques of existing MVC approaches [3, 6, 11, 19], where Chao et al. [11] reviewed generative and discriminative approaches, elaborated on the relationships between MVC and several closely related learning approaches (multi-view representation, ensemble clustering, multi-view supervised and semi-supervised learning, and multi-task clustering), and introduced several real-world applications of MVC. Yang and Wang [3] classified MVC approaches into five categories (i.e., co-training style approaches, multi-kernel learning, multi-view graph clustering, multi-view subspace clustering, and multi-task MVC), provided a few examples for each category of approaches, and listed some publicly available multi-view datasets. Fu et al. [6] summarized graph-based approaches, space learning based approaches, and binary code learning based approaches. Wen et al. [19] reviewed the existing studies on approaches for IMVC, categorizing them into MF-based IMVC, kernel learning based IMVC, graph learning based IMVC, and deep learning based IMVC. The study then selected representative IMVC approaches for an experimental comparative analysis. Several authors [3, 6, 11, 19] also examined various open problems that may necessitate further investigation. These include challenges related to large-scale data (size and dimension), complex data with noises or mixed types, imbalance information, missing view/value recovery, local minima, and deep learning, among others.

However, these survey studies mainly focus on approaches for complete multi-view data (where each feature is collected and each sample appears in each view), or those for incomplete multi-view data with missing samples or features. They do not explore approaches designed to handle both complete and incomplete data at the same time. Moreover, these studies tend to overlook approaches for uncertain multi-view data and dynamic multi-view data. In addition, these survey studies predominantly focus on approaches proposed before 2019. Many novel and important approaches developed after that, such as deep learning based approaches [20, 21], have not been considered by existing survey studies. This gap makes it challenging to comprehensively track the current progress of the field.

Furthermore, among these survey studies, only Fu et al. [6] and Wen et al. [19] provided experimental quantitative evaluations. In the work of Fu et al. [6], eight approaches (one k-means, three graph based, three space learning based, and one binary code learning based) were tested on seven complete multi-view datasets. Wen et al. [19] executed evaluations for 17 approaches (two single view, eight MF based, one adaptive neighbors based, two kernel learning based, and four graph based) on five incomplete multi-view datasets. Neither Fu et al. [6] nor Wen et al. [19] conducted experimental evaluations for the deep learning based approaches. As a result, although a number of MVC approaches have been proposed, and several reviews assessing these approaches have been conducted, there is still a lack of comprehensive review and quantitative measurement and comparisons of MVC approaches. Especially noteworthy is the lack of evaluation for the approaches developed after 2019.

Hence, there is a need for a thorough review and quantitative assessment of MVC approaches, particularly those employing deep learning techniques. Notably, these crucial approaches have been overlooked in existing survey studies. Thus, this article focuses on investigating MVC approaches, including those proposed after 2019, which have not been reviewed by existing survey studies. Additionally, this work conducts an experiment-based evaluation study for the most representative MVC approaches. The aim is to provide a quantified assessment of representative MVC approaches. This assessment will provide researchers with a comprehensive understanding of various approaches while offering practitioners measurable insights to guide their selection of suitable methods for specific applications.

To achieve this objective, we first analyze and formalize basic concepts by introducing common key techniques for MVC. This includes the introduction of multi-view data, the definition of the MVC problem, principles related to MVC, strategies for fusing information and weighting views, the MVC routine, and the model structure, as well as the optimization scheme. Subsequently, we propose a novel taxonomy of MVC approaches and present the characteristics of representative MVC approaches. The MVC approaches are classified into four categories: complete MVC, IMVC, uncertain MVC, and dynamic MVC, based on the data types they handle. Additionally, the approaches for complete MVC and IMVC are further categorized into eight and six sub-categories, respectively, based on their adopted working mechanisms and techniques.

Moreover, we summarize commonly used datasets and performance metrics for evaluating clustering performance in the field of MVC. The representative MVC datasets are divided into five categories: text, image, text-gene, image-text, and video datasets. The performance metrics include internal indices, which evaluate a clustering algorithm by summarizing results in a single quality score, and external indices, which evaluate the clustering result by comparing it with externally supplied true labels.

Finally, we conduct an empirical evaluation on 35 representative MVC approaches using seven real-world benchmark datasets. The 35 MVC approaches selected for our experimental evaluation studies include 29 complete MVC approaches and 6 IMVC approaches. These approaches cover the main categories in the taxonomy proposed in this article, each characterized by distinct structures or constraints.

The seven datasets utilized in our study, each with varying scales, comprise three text datasets, three image datasets, and one image-text dataset. These datasets are widely recognized as typical in the MVC communities. Additionally, these seven datasets, serving as well-known benchmarks, represent different typical application scenarios, each with distinct views, classes, instances, features, and feature dimensions.

The clustering approaches and datasets have been meticulously selected to investigate the clustering performance of various approaches, validate the factors affecting clustering performance, and test the ability of various approaches to handle different data scales, ensuring the validity and relevance of this evaluation study. Furthermore, to support researchers and practitioners conveniently, we have compiled related references, implementations, and datasets on GitHub.[1]

Our contributions are summarized as follows:

- We formalize and analyze basic concepts and common key techniques for MVC, which provides the background knowledge for understanding MVC and its related issues.
- We propose a novel taxonomy of MVC approaches based on the detailed analysis of existing approaches, which offers a comprehensive picture of MVC approaches developed.
- We provide a critical review on the working mechanisms and characteristics of the representative MVC approaches, and summarize representative MVC datasets and performance metrics commonly used in the MVC assessment and evaluation.
- We present the current progress of MVC, addressing the existing gap that the latest approaches proposed after 2019, particularly the deep (contrastive) learning based MVC approaches, have not been evaluated in existing survey studies.

We selected 35 representative MVC approaches based on the proposed taxonomy to conduct an empirical evaluation on seven real-world benchmark datasets, exploring how they perform across different types of datasets. The experimental results reveal that most MVC approaches struggle

[1]https://github.com/dugzzuli/A-Survey-of-Multi-view-Clustering-Approaches

with large-scale datasets. No MVC approach consistently maintains high performance across all types of datasets. Factors such as model structures, regularization constraints, and weights corresponding to different views contribute to improving clustering performance. For example, the deep learning based approach with multi-level representations and adversarial regularization performs well across many datasets. These findings provide valuable insights for future practitioners, offering information on the performance features of existing approaches. This serves as a practical guide for the development of applications, providing empirical evidence for selecting suitable approaches in specific circumstances.

The rest of the article is organized as follows. In Section 2, we analyze and introduce the basic concepts and some common strategies used in MVC approaches. Following that, we provide a novel taxonomy of MVC approaches. In Sections 3 through 6, we will present the working mechanisms and characteristics of representative MVC approaches proposed in recent years, each corresponding to the categories outlined in the proposed taxonomy. Afterward, we review the datasets and performance metrics used in the literature on MVC evaluation in Sections 7 and 8, respectively. We then present the empirical results and analyses in Section 9. Finally, we discuss potential directions for future research in Section 10 and conclude this study in Section 11.

2 PRELIMINARIES

In this section, we introduce categories of multi-view data, the definition of MVC, two principles of MVC, common strategies for solving MVC, and propose a new taxonomy of MVC approaches.

2.1 Multi-View Data

Multi-view data refers to the same sample is described from various perspectives, with each perspective capturing a class of features known as a view [4, 22]. Figure 1 illustrates four intuitive examples of multi-view data, showing a person's iris, fingerprint, and face (see Figure 1(a)); a multi-wavelength whirlpool galaxy (see Figure 1(b)); car images taken from different viewpoints (see Figure 1(c)); and "thank you" described in many languages (see Figure 1(d)). Although these different features may have distinct physical meanings, take different forms, and exhibit heterogeneous and diverse properties, they all represent the same sample from different perspectives [3, 23]. Compared to single-view data that describes samples from a single perspective, multi-view data is semantically richer, more informative, and diverse but also more complex to analyze.

Multi-view data can be categorized as either complete or incomplete, depending on whether all expected perspectives or views are available for analysis. In incomplete multi-view data, only partial features could be available in some views for some data samples (missing features/values), or some data samples could be missing their whole observations in some views (i.e., missing samples/views), or even all samples of a cluster are not observed in one view (missing clusters) [24, 25]. Missing cluster and missing sample can be considered as one special form of missing feature. Figure 2 illustrates an example of incomplete multi-view data. In the figure, samples within the same cluster are represented by the same shape, distinguished by color. The dotted shape indicates missing samples, and "NA" represents missing feature values. The incomplete multi-view data is ubiquitous in the real world due to sensor failure, equipment malfunction, data corruption, and other factors.

Multi-view data may also be certain or uncertain. Certain multi-view data implies that the features of each sample and their values are affirmative, whereas uncertain multi-view data refers to the data that deviates from the desired distribution or pattern due to randomness, privacy issue, and imprecision in measurement [26]. In real-world applications, data collected from various sources like smartphones, satellites, and transportation typically exhibit uncertainty, often quantified through probability distributions.

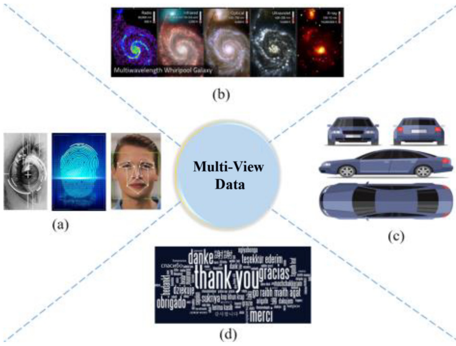

Fig. 1. Examples of multi-view data.

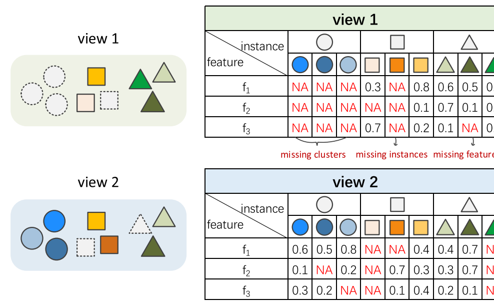

Fig. 2. An example of incomplete multi-view data.

In addition, in many practical applications, the number of views dynamically changes over time rather than remaining fixed. For example, in brain-computer interface systems, the acquisition of brain signals plays a crucial role in analyzing brain states. Since the brain signal changes with the object's mental state, it becomes necessary to collect the signal at different times. Therefore, the signal data at each time serves as a view, and the number of views changes dynamically over time [27].

2.2 The Problem Definition

Let $\boldsymbol{X} = \{\boldsymbol{X}^1, \boldsymbol{X}^2, \ldots, \boldsymbol{X}^m\}$ be a dataset with m views and $\boldsymbol{X}^v = \{\boldsymbol{x}_1^v, \ldots, \boldsymbol{x}_n^v\} \in \Re^{d_v \times n}$ be the $v-$th view data, where d_v is the feature dimensionality of the $v-$th view, n is the number of data samples, and $\boldsymbol{x}_i^v$ represents the feature vector of the $i-$th sample in the $v-$th view.

Let $C = \{C_1, C_2, \ldots, C_K\}$ be a set of K clusters, called a *cluster structure*, where C_j is the set of $|C_j|$ samples assigned to the $j-$th cluster, $\sum_j |C_j| = n$. Let $\boldsymbol{Y} \in \Re^{n \times K}$ be a membership matrix, whose $(i, j)-$th entry y_{ij} represents the probability that the $i-$thsample belongs to the $j-$th cluster. The sum of each row entries of Y should be 1 to make sure each row is a probability—that is, $\sum_j y_{ij} = 1$. If only one entry of each row is 1 and all others are 0, it is a hard clustering (each sample can only be assigned to one specific cluster with the probability of 1); otherwise, it is a soft clustering (each sample is assigned to one cluster with some probability between 0 and 1).

Given $\boldsymbol{X} = \{\boldsymbol{X}^1, \boldsymbol{X}^2, \ldots, \boldsymbol{X}^m\}$, the purpose of MVC is to divide n samples into K clusters by integrating the heterogeneous features of $\boldsymbol{X}$ without any label information such that data samples within the same cluster are more similar than those in different clusters. In other words, finally we will get a membership matrix $\boldsymbol{Y}$ or a cluster structure $C = \{C_1, C_2, \ldots, C_K\}$ to indicate which samples are in the same group while others are in other clusters.

The key questions in MVC include (1) how to effectively utilize the otherness information among different views in a multi-view dataset? (2) how to completely discover the consistency information between the different views? (3) how to reduce computation and space complexities? (4) how to enhance the robustness against noise and outliers? and (5) how to prevent being trapped in sub-optimal local minima/maxima? Both the otherness and consistency information in multi-view data are quite useful for effective clustering analyses. Multi-view data usually lies in high-dimensional space, where redundant and irrelevant features may result in the curse of dimensionality. Meanwhile, data often contain noise and outliers, which may destroy the underlying clustering structure. Moreover, approaches for solving MVC are usually non-convex, making them prone to becoming stuck into sub-optimal local minima, especially when there are outliers and missing data.

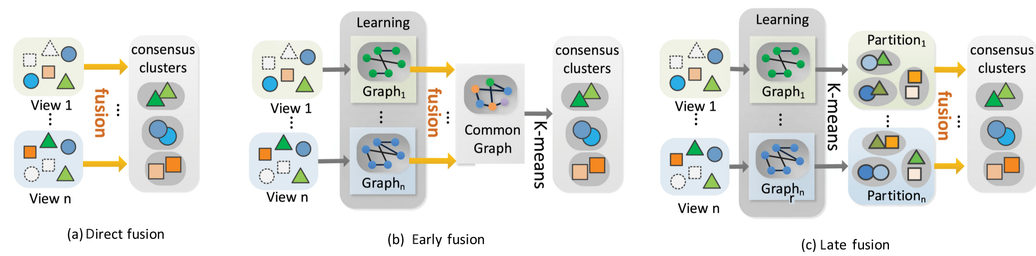

Fig. 3. An illustration of three fusion strategies.

2.3 Principles Related to MVC

There are two significant principles ensuring the effectiveness of MVC: consensus and complementary principles [4]. The context of multi-view data means that there is some common knowledge across different views (e.g., both two pictures about dogs have contour and facial features), whereas the complementary of multi-view data refers to some unique knowledge contained in each view that is not available in other views (e.g., one view shows the side of a dog and the other shows the front of the dog, and these two views allow for a more complete depiction of the dog). Therefore, the consensus principle aims to maximize the agreement across multiple distinct views for improving the understanding of the commonness of the observed samples, whereas the complementary principle states that in a multi-view context, each view of the data may contain some particular knowledge that other views do not have, and this particular knowledge can be mutually complementary.

Both complementary and consensus principles play important roles for improving the performance of MVL algorithms [28]. By exploring the consistency and complementary properties of different views, MVL is rendered more effective and promising, and exhibits better generalization ability than single-view learning [4].

2.4 The Information Fusion Strategy

The strategies for integrating information from multiple views can be divided into three categories—direct fusion, early fusion, and late fusion—based on the fusion stage. They are also referred to as data-level, feature-level, and decision-level fusion, respectively (i.e., fusion in the data, fusion in the projected features, and fusion in the results). Direct fusion approaches involve the direct incorporation of multi-view data into the clustering process by optimizing specific loss functions. Early fusion combines multiple features or graph structure representations of multi-view data into a single representation or a consensus affinity graph across multiple views. Subsequently, any well-known single-view clustering algorithm, such as k-means, can be applied to partition data samples. Most approaches learn a graph structure representation for each view by deploying features of different views, then a consensus affinity graph is built. Some other approaches directly learn a common graph matrix from the original feature space. In contrast, the approaches of the late fusion first perform data clustering on each view individually and subsequently fuse the results for all views to obtain the final clustering results through consensus [29]. The advantage of late fusion is that it reduces the interference of other information channels to every separate partition such that the effect of random noise can be reduced. Figure 3 illustrates the three fusion strategies. There is no theoretical foundation to decide which one is the best.

2.5 The Clustering Routine

There are two clustering routines—one-step routine and two-step routine—to execute MVC. The two-step routine first extracts the low-dimensional representation of multi-view data and then uses traditional clustering approaches, such as k-means, to process the obtained representation. In other words, the two-step routine often needs a post-processing process, such as applying a simple clustering method to the learned representation or conducting a fusion operation on the clustering results of individual views, to produce the final clustering results. This two-step learning strategy may lead to unsatisfactory clustering performance since the learning of the multi-view representation is not informed by the final clustering goal. The correlation between these two steps is not fully explored, potentially making the learned low-dimensional representation unsuitable for subsequent clustering tasks. On the contrary, the one-step routine integrates representation learning and clustering task into a unified framework, simultaneously learning a graph for each view, a partition for each view, and a consensus partition. Based on an iterative optimization strategy, high-quality consensus clustering results can be obtained directly and employed to guide the graph construction and the updating of basic partitions. This, in turn, contributes to the formation of a new consensus partition. Through joint optimization, co-training involves simultaneous clustering and representation learning, leveraging the inherent interactions between two tasks and realizing the mutual benefit of these two steps [1]. In the one-step routine, the cluster label of each data sample can be directly assigned and without the need for any post-processing, reducing the instability of the clustering performance caused by the uncertainty of post-processing operations.

2.6 The Weighting Strategy

In MVC, how to merge multiple views is a key issue. It is essential to assign an appropriate weight to each view based on its importance for the merging process. This ensures a deeper exploration of the complementary information present in heterogeneous data and effectively reduces the adverse effects of noise and outliers. The simplest approach is to assign the same weight to all views (equal weighted) or not to consider the weight, treating all views equally and assuming their reliability. However, the clustering results may be degraded if different views are not distinguished [30]. In real-world clustering problems, data views inherently vary in strength, and each view possesses specific statistical properties while being susceptible to different forms of noise pollution. Thus, it is unreasonable to treat different views equally during the clustering process. To distinguish the different contributions of different views, auto-weighted (self-weighted) approaches, which involve automatically learning the weight of each view [31], have been proposed. Auto-weighted approaches can be classified into two categories: parameter weighted and parameter-free weighted strategies. Parameter-weighted approaches learn the weight of each view by introducing additional hyperparameters, which control the smoothness or sparsity of the weight distribution. However, setting these hyperparameters in the clustering task is often challenging due to the need for an extensive search in a large parameter space. The clustering results can be sensitive to these hyperparameters, and the optimal values may vary across different datasets. The parameter-free weighted strategy automatically assigns weights by a self-conducted weight learning, without the requirement of any hyperparameters while maintaining precision [32].

Except for weighting the view, Du et al. [33] learned the weights of different features in each view and the weights of each sample in different views by introducing a feature-level and a sample-level attention mechanism. This approach hierarchically distinguishes the weights of different features in one view and the weights of the same sample in different views. Using a unique weighting strategy, Zhang et al. [34] considered the confidence of both views and samples under the assumption that samples may have different confidence levels under the same view. Xu et al. [35] assigned weights to the views of data samples and feature representations in each view, emphasizing discriminatory features and views over others. Liu et al. [36] learned the cluster-wise weights instead of view-wise weights, with the cluster-weighted scheme enhancing the interpretability of the clustering results. Hu et al. [37] automatically learned the view weights based on the concept of mutual information and then imposed them simultaneously on the content-based and context-based multi-view data representations. Kang et al. [38] measured the weights of the previous views and the last view when the number of views increases over time.

2.7 The Model Structure

The MVC models consist of shallow structure or deep structure. Models with shallow structure learn the low-dimensional representation of multi-view data via a one-level structure, ignoring the hierarchical and non-linear structural information hidden in each view. For example, classical NMF only factorizes the data matrix $\boldsymbol{X}$ into two non-negative factor matrices $\boldsymbol{U}$ and $\boldsymbol{V}$ such that $\boldsymbol{X} \approx \boldsymbol{UV}$, which may limit its ability to learn higher-level and more complex hierarchical information. Models with deep structure learn a low-dimensional representation of multi-view data via a multi-level structure, enabling them to capture complex hierarchical and structural information. For example, the deep semi-NMF model [39] factorized data matrix $\boldsymbol{X}$ into $l+1$ factors $\boldsymbol{X}^{\pm} \approx \boldsymbol{U}_1^{\pm}\boldsymbol{U}_2^{\pm}\cdots\boldsymbol{U}_l^{\pm}\boldsymbol{V}_l^{+}$, which allows for a hierarchy of l layers of implicit data representations that can be given by the following factorizations: $\boldsymbol{V}_{l-1}^{+} \approx \boldsymbol{U}_l^{\pm}\boldsymbol{V}_l^{+}, \ldots, \boldsymbol{V}_2^{+} \approx \boldsymbol{U}_3^{\pm}\cdots\boldsymbol{U}_l^{\pm}\boldsymbol{V}_l^{+}, \boldsymbol{V}_1^{+} \approx \boldsymbol{U}_2^{\pm}\cdots\boldsymbol{U}_l^{\pm}\boldsymbol{V}_l^{+}$.

2.8 The Optimization Scheme

In general, MVC is an NP-hard optimization problem. The most commonly used solution to this problem is the alternating iterative optimization scheme, which decomposes the problem into several tractable sub-problems. Each variable or group of variables is updated alternately while keeping the others fixed. For example, Xu et al. [40] solved the deep multi-view concept learning (DMCL) model using block coordinate descent [41] that each time optimizes one group of variables while keeping the other groups fixed. The augmented Lagrange multiplier [42] and alternating direction method of multipliers (ADMM) [43] are widely used to solve the convex optimization problem.

2.9 The Proposed Taxonomy

In this article, we propose a novel taxonomy according to the data types that MVC approaches deal with. This taxonomy encompasses approaches for complete, incomplete, uncertain, and dynamic multi-view data, denoted as complete MVC, IMVC, uncertain MVC, and dynamic MVC for brevity. According to the working mechanisms and techniques that various approaches adopted, the approaches for complete multi-view data are further divided into eight sub-categories: (1) NMF-based approaches, (2) **Multiple Kernel Learning (MKL)**-based approaches, (3) graph-based approaches, (4) subspace-based approaches, (5) deep learning based approaches, (6) contrastive learning based approaches, (7) co-learning-based approaches, and (8) self-paced learning based approaches. Similarly, the approaches for incomplete multi-view data are further divided into six sub-categories: (1) NMF-based approaches, (2) MKL-based approaches, (3) graph-based approaches, (4) subspace-based approaches, (5) deep learning based approaches, and (6) contrastive learning based approaches. The approaches for uncertain multi-view data and those for dynamic multi-view data are not further classified into sub-categories, as there are not many approaches proposed in these two categories. The proposed taxonomy of MVC approaches is illustrated in Figure 4.

The mechanisms and principles of the representative MVC approaches proposed in recent years are summarized in the next four sections. Note that, due to the space limitation, the tables summarizing the characteristics of various approaches are placed in the appendix.

3 COMPLETE MVC

In this section, we present the working mechanisms of the representative MVC approaches for complete multi-view data and their characteristics. The general procedure of the approaches for

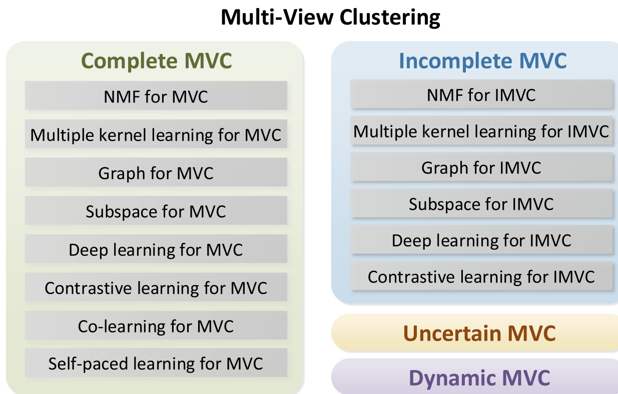

Fig. 4. The proposed taxonomy of MVC approaches.

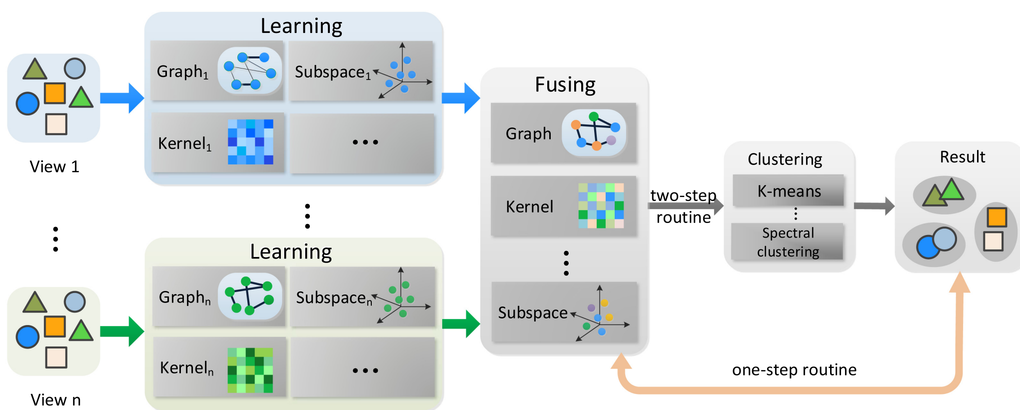

Fig. 5. The general procedure of complete MVC approaches.

complete MVC is shown in Figure 5, whereas Appendix Table 1 summarizes the characteristics of the representative complete MVC approaches. These characteristics include motivation, model structure, information fusion and weighting strategy, clustering routine, model peculiarity, and time as well as space complexity.

3.1 NMF-Based Approaches for MVC

NMF-based MVC approaches decompose a data matrix with non-negative elements into two low-rank matrices. The product of these two low-rank matrices is designed to approximate the original data matrix. The non-negative constraint results in a parts-based representation of samples, which accords with the cognitive process of the human brain from psychological and physiological evidence. This characteristic makes the clustering results easy to interpret.

To obtain the desired dimensional-reduced representation, various constraints are integrated into traditional NMF. For example, Liu et al. [16] introduced the consensus constraint to push a clustering solution of each view toward a common consensus. Additionally, Wang et al. [44], Zong et al. [45], and Liang et al. [46] imposed multi-manifold regularization to preserve the locally geometrical structure of the data space; Cai et al. [47] utilized the sparseness constraint to extract the robust feature of each view; and Cai et al. [48] and Liang et al. [49] designed the orthogonality constraint to capture the intra-view diversity, in turn, to obtain the desirable representations for each view. Zheng et al. [50] exerted a graph Laplacian regularization on the indicator matrix learned via matrix factorization assisted k-means. This approach aims to capture the intrinsic geometric structure of original data. Luo et al. [51] preserved the geometric structures of multi-view data in both the data space and the feature space. Liu et al. [52] adopted auto-weighted collective matrix factorization (CMF) to extract shared information of multi-view data. Additionally, the approach imposed graph dual regularization terms with orthogonality constraints to preserve the geometrical structure of the decomposed factors.

To reduce the high dependency on the quality of the original views and recognize global relationships among data samples, Zhou et al. [53] jointly factorized multiple networks transformed from multi-view data. The approach also incorporated sparse as well as multi-manifold regularization into NMF to keep the intrinsic geometrical information of the multi-view network manifold space.

To accelerate computational efficiency and decrease memory costs, Yang et al. [21] employed NMF to the embedded anchor graph, and utilized correntropy to increase clustering robustness. In another work, Yang et al. [54] divided the optimization problem into three decoupled small-scale problems containing only a small amount of matrix multiplications. Houfar et al. [55] exploited a constrained binary matrix factorization to achieve direct clustering. Zhang et al. [56] encoded the multi-view image descriptors into a compact common binary code space and clustered the collaborative binary representations. This process is designed to reduce computation costs and storage requirements through bit operations. Zhao et al. [57] developed an orthogonal mapping binary graph approach to eliminate redundant information and extract local geometric structure information of binary codes.

To capture the complex hierarchical and non-linear information, Chang et al. [58] introduced deep concept factorization (CF) into MVC. Zhang et al. [59] utilized deep matrix decomposition to obtain the partition matrix of each view. Li et al. [60] constructed a multi-layer NMF model with graph regularization to extract abstract representations. The last layer representation from each view was then derived to form a common consensus representation. Du et al. [61] designed multiple encoder components and decoder components with deep structures to hierarchically factorize the input data. All encoder and decoder components were then integrated at an abstract level to capture heterogeneous information across multi-view data. Chen et al. [62] designed diversity embedding deep matrix factorization to obtain discriminative features and reduce feature redundancy. Zhao et al. [63] adopted semi-NMF to learn the hierarchical semantics of multi-view data in a layer-wise fashion. In this process, the non-negative representation of each view in the final layer was enforced to be the same, maximizing the mutual information from each view. Xu et al. [40] performed hierarchically non-negative factorization for capturing semantic structures. Additionally, the approach explicitly modeled both consistency and complementary information at the highest abstraction level.

3.2 MKL-Based Approaches for MVC

MKL-based approaches linearly or non-linearly combine pre-defined kernels corresponding to different views to improve clustering performance. These pre-defined kernels map samples from the original low-dimensional space to a high-dimensional space such that the samples are linearly separable in high-dimensional space. Due to the effectiveness of handling non-linear data and avoiding the selection of specific kernel function (in general, it is very time consuming and expensive to select the most suitable kernel from a pre-specified pool of base kernels, e.g., linear kernel, polynomial kernel, and Gaussian kernel), MKL-based approaches for MVC have been widely investigated and have achieved promising results [64].

Ren et al. [65] and Huang et al. [66] learned similarity relationships in kernel spaces to improve the robustness against noise. Chen et al. [67] jointly learned a kernel representation tensor and affinity matrix. Zhang et al. [68] employed kernel-induced functions to map the multi-view data from linear space into non-linear space for utilizing the non-linear structure hidden in data. These approaches automatically appointed a reasonable weight for each view.

3.3 Graph-Based Approaches for MVC

The graph is an important data structure for representing the relationships hidden in multi-view data. Each node in a graph corresponds to a data sample, and each edge represents a relationship between two samples. Graph-based clustering approaches seek to learn a consensus affinity graph across all views and then carry out graph-cut algorithms or other techniques (e.g., spectral clustering) on the consensus affinity graph to produce the clustering result. The consensus affinity graphs can be directly learned in original feature space, or obtained by fusing multiple candidate graphs learned in original feature space separately. It is obvious that the clustering performance highly depends on the quality of the consensus affinity graph, thus it is a crucial task to learn a high-quality consensus affinity graph. Usually, the differences among various multi-view graph clustering approaches vary in how they learn the consensus affinity graph across the multi-view data.

To obtain a high-quality consensus affinity graph, some works [20, 69–72] projected the multi-view data into a low-dimension shared latent embedding space. They then learned the consensus affinity graph from the shared latent embedding space to reduce the corruption and redundancy of the original views. Jing et al. [73], Wang et al. [74], and Wang et al. [75] employed manifold learning and sparse representation to construct the graph of each view and fused them in an automatically weighted way. Xu et al. [76], Zhao et al. [77], and Chen et al. [78] introduced tensor nuclear norm to minimize the divergence between graphs of different views. Hajjar et al. [79] introduced a consistent smoothness constraint of overall views and an orthogonality constraint. Wang et al. [80] converted multi-view fuzzy clustering to adaptive graph learning with sparse low-rank constraints to ensure its strong discriminative ability. Chen et al. [81] weighted the different views in terms of their confidence.

To reduce the computational complexity, Chen et al. [82] relaxed the constraint of the global similarity matrix. Alshammari et al. [83] devised a refined version of the k-nearest neighbor graph to keep data points and aggressively reduced the number of edges. Li et al. [84] and Hu et al. [85] developed MM (Majorization-Minimization)-based optimization approaches. Shi et al. [86] and Zhong et al. [87] jointly optimize the learning of consensus graph and discretization of cluster labels. Wang et al. [88] learned a structured graph to directly extract the clustering indicators, without performing other discretization procedures. Kang et al. [89] constructed a bipartite graph to depict the relationship between samples and anchor points, and imposed a connectivity constraint to guarantee that the connected components indicate clusters directly.

3.4 Subspace-Based Approaches for MVC

Subspace-based approaches for MVC aim to directly learn a unified low-dimensional representation shared by multiple views from multiple subspaces or a pre-learned latent space. This is based on the assumption that the multi-view observations are generated from an underlying latent representation [90]. Subsequently, k-means or spectral clustering is applied to the unified representation to divide data samples lying in a union of multiple low-dimensional subspaces into different clusters. This process ensures that samples in the same cluster come from one subspace. The key challenges for subspace-based MVC approaches include learning a robust representation, addressing data with non-linear structures, and enhancing computation efficiency.

To learn a robust representation, Si et al. [91] employed an exclusivity constraint term to enhance the diversity of specific representations among different views. Additionally, the approach imposed a clustering structure constraint on the learned subspace self-representation to obtain a clustering-oriented subspace self-representation. Qin et al. [92] built an anti-block diagonal indicator matrix, incorporating a small amount of supervisory information. This was done to regularize the shared affinity matrix corresponding to the latent representation for ensuring its block diagonal structure. Liu et al. [93] defined a local graph regularization term about the consensus latent subspace representation. This term was designed to preserve the manifold structure of data and ensure consistency across different views. Wang et al. [94] employed the maximum dependence constraint between the similarity matrix and its latent intact (complete and not damaged) points to build an similarity matrix. Yang et al. [95] designed the multiplicative decomposition scheme and the variable splitting scheme to extract the components from their corresponding view-specific coefficients, where inconsistent elements are filtered out. Kang et al. [1] fused multi-view information in a partition space to reduce the effect of noise. Cai et al. [96] adopted the view-consensus grouping effect and low-rank constraint via the nuclear norm to regularize the view-commonness representation. Zhou et al. [97] captured the correlations between shared information across multiple views and employed view-specific information to describe specific property of each independent view. Cao et al. [98] adopted the weighted nuclear norm (instead of nuclear norm) to approximate the rank of the common coefficient matrix. Lu et al. [99] hired the self-attention mechanism to derive dynamic weights of different views for fusing consistent and view-specific information from multiple views.

To capture the non-linear nature of data, Zhang et al. [100] and Zhang et al. [101] exploited a low-rank kernel mapping and the non-convex Schatten p-norm regularizer as well as the correntropy to learn a joint subspace representation of all views. Zhang et al. [102], Fu et al. [103], and Wang et al. [104] projected data from original space into a kernel/tensor space to encode the low-rank property of the self-representation tensor. Rong et al. [105] merged the self-representative subspaces of different views on a Grassmann manifold. Chen et al. [106] and Li et al. [107] utilized a kernel trick and kernel dependence measure to encode complementary information from different views. Xue et al. [108] employed deep matrix factorization to obtain multi-view multi-layer low-rank subspace representations. Zhu et al. [109], Yang et al. [110], Wang et al. [111], and Si et al. [112] employed the deep convolutional/auto-encoder to learn the latent low-dimensional hierarchical representation. Zheng et al. [113] and Du et al. [114] employed auto-encoder networks on multiple views to achieve multi-level latent smoothness.

To improve the computation efficiency, Tao et al. [90] directly recovered the row space of the latent representation without the graph construction procedure. Wang et al. [115] jointly optimize the anchor selection and subspace graph construction in a parameter-free way.

3.5 Deep Learning Based Approaches for MVC

Deep learning based MVC approaches utilize neural networks to learn latent representations, then divide the data sample into different groups based on the learned latent representations. Because the learned latent representations can reveal the potential peculiarities of complementarity and consistency information among multi-view features, such as the mutual agreement, non-linear relationships, they effectively mitigate the effect of noise and dimensionality curse.

Li et al. [116] utilized auto-encoders to learn latent representations shared by multiple views, and leveraged adversarial training to further capture the data distribution and disentangle the latent space. Du et al. [117], Xu et al. [118], and Wang et al. [119] used auto-encoders to learn individually the embedded representations of multiple views with the consideration of both consensus and complementarity of multiple views. Xie et al. [120] employed various auto-encoders, such as stacked auto-encoder (SAE), convolutional auto-encoder (CAE), and convolutional variational auto-encoder (Conv-VAE), to exact multi-view features.

To encode graph structure and node attribute to the node representation, Xia et al. [121] utilized a **Graph Convolutional Network (GCN)** with a block diagonal property to encode the discriminative information. Additionally, they utilized Euler transform to augment the node attribute as a new view descriptor for non-Euclidean structure data. Xiao et al. [122] adopted a graph neural network (GNN) to implement dual fusion-propagation for capturing the multiple information among different views. Wang et al. [123] built a neural network composed of multiple blocks to learn sparse regularizers.

To distinguish the importance of different views and semantics, Zhou and Shen [124] adopted adversarial learning and the attention mechanism to align the latent feature distributions of different views and quantified the importance of modalities. Huang et al. [125] adopted the self-attention mechanism to calculate the alignment matrix for capturing the category-level correspondence of the unaligned data. Fang et al. [126] developed a differentiable bi-level optimization network to enhance the interpretability of deep MVC.

3.6 Contrastive Learning Based Approaches for MVC

Recently, contrastive learning, aiming at maximizing the agreement between positive sample pairs and minimizing that between negative sample pairs, has been widely used in MVC. This is attributed to its excellent ability to capture the consistency of multiple views. It offers a way to align representations from different views at the sample level, forcing the label distributions to be aligned as well [127].

Fang et al. [128] adopted contrastive learning to infer the inter-cluster relationships and intra-cluster boundaries from the local context of each node. Du et al. [129] used contrastive learning to train the GCN for integrating the topological structures and node features of neighbors. Lin et al. [130] used contrastive learning to align sample-level representations across multiple views for capturing the view-invariance information. Xu et al. [131] used contrastive learning to achieve the consistency objectives for the high-level features and the semantic labels. Liu et al. [132] respectively exploited a cross-view contrastive learning and a mutual contrastive teacher-student learning. These approaches aimed to obtain a redundancy-free consistent representation at the instance level and capture the intra-view discriminative information at the semantic level. Deng et al. [133] designed instance-level and cluster-level contrastive learning to exploit the representations of the augmented weak-weak view pair and the strong-weak view pairs.

To improve performance, Wang et al. [134] developed a contrastive fine modeling via maximizing the similarity of pair view to guarantee the consistency of multiple views. Pan and Kang [135] designed a graph contrastive loss to regularize the learning of the consensus graph. Wang et al. [136] introduced a contrastive reconstruction loss to realize sample-level approximations between the reconstructed graphs and the raw graphs. However, Trosten et al. [137] proved that contrastive alignment can be detrimental to the clustering performance, especially when the number of views increases.

3.7 Co-Learning-Based Approaches for MVC

Co-learning approaches aim to improve the clustering performance by exchanging information among different objects. Co-learning approaches include co-regularization MVC, co-training MVC, multi-task MVC, and co-clustering MVC. The general procedures of co-training, multi-task, and co-clustering MVC are shown in Figure 6.

The typical representatives of co-regularization MVC and co-training MVC are CoRegSC (co-regularization spectral clustering) [15] and CoTrainSC (co-training spectral clustering) [138].

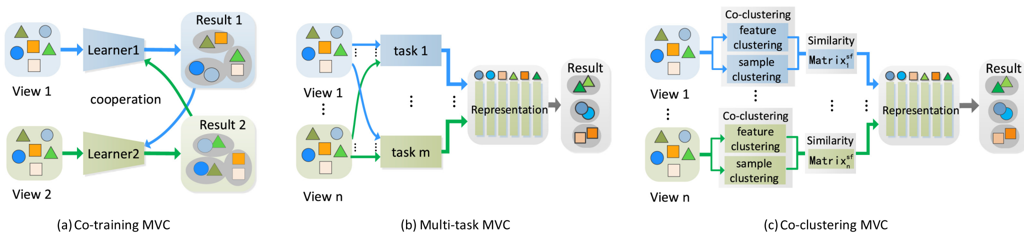

Fig. 6. General procedures of co-training, multi-task, and co-clustering MVC.

CoRegSC formally measured the agreement on distinct views and solved the corresponding objective problem to make the cluster structures in different views agree with each other. CoTrainSC employed separate but correlated learners on each view to acquire the cluster structure. The acquired cluster structure then guided the learners in other views, facilitating the exchange of information among different views. This allowed the disagreement among views to be propagated back to the learners, helping them learn a more accurate cluster structure by minimizing the disagreement in the next iteration. Through an iterative alternate training procedure, the cluster structures of multiple views tended toward consensus. Blum and Mitchell [139] pointed out that the success of co-training algorithms mainly relies on three assumptions: (1) *sufficiency*, where each view is sufficient for classification on its own; (2) *compatibility*, where the target functions of both views predict the same labels for co-occurring features with a high probability; and (3) *conditional independence*, where views are conditionally independent given the label. Deng et al. [140] implemented collaborative learning between visible and hidden views to combine the individual information and the shared information in different views.

Multi-task clustering improves individual clustering performance by learning the relationship among related tasks, whereas MVC makes use of consistency among different views to achieve better performance. Multi-task MVC means the tasks are closely related and each task can be analyzed from multiple views. The goal of multi-task MVC is to generate a consensus partitioning for every task by exploiting the shared relationships between the views within a task and between different tasks [141]. Mitra and Saha [141] formulated the multi-task MVC problem as a multi-objective optimization problem. Zhang et al. [142] and Zhang et al. [143] developed a semi-non-negative matrix tri-factorization-based multi-task MVC clustering algorithm to deal with the data with negative feature values.

Co-clustering refers to conducting two-sided clustering along the samples and features simultaneously under the assumption that samples exhibit a pattern only under a subset of features [14]. Co-clustering MVC combines MVC with co-clustering for taking advantage of the diversity of features provided by multiple sources and the dual relationship between the feature space and sample space.

To employ the agreement and disagreement among views, Xue et al. [144] shared a common clustering results along the sample dimension and kept the clustering results of each view specific along the feature dimension. The mechanism of maximum entropy was leveraged to control the importance of different views. Huang et al. [145] built a view-level bipartite graph to draw the co-occurring structure of data for exploiting the duality between samples and features of multi-view data.

To improve the robustness against noisy features, Hu et al. [146] employed the dynamic multi-view co-clustering algorithm with mutual information. This approach was employed to learn the view weights, which was then applied to the discriminative feature representations of multiple views, instead of the original representations. Zhao et al. [147] employed multiple co-clustering algorithms to calculate the sample-sample, feature-feature, and sample-feature similarity information of each view data. Hussain et al. [14] used co-clustering as the basic clustering block of MVC, and utilized the features and view weighting schemes to iteratively specify the perceived features and view importance.

In addition, Nie et al. [148] conducted co-clustering based on matrix factorization under the constraints of an indicator matrix to improve the computational efficiency. Du et al. [149] developed a differentiable deep network to learn interpretable and consistent collaborative representation from multi-source features, and maintain sparsity between multi-view feature space and single-view sample space.

3.8 Self-Paced Learning Based Approaches for MVC

Self-paced learning, an effective technique to avoid bad local minima and improve the generalization result, simulates the human learning process. It starts by training a model on 'easy' examples which have smaller loss values and then gradually takes 'complex' examples into consideration. The general self-paced learning model is composed of a weighted loss term on all examples (with higher losses) and a regularizer term imposed on example weights. By gradually increasing the penalty on the regularizer during model optimization, more examples are automatically included in training from 'easy' to 'complex' via a pure self-paced approach.

Chen et al. [106] employed a low-rank tensor constraint to assign different weights on different singular values of the representations and used the self-paced learning to treat multiple instances differently. Ren et al. [150] exploited a self-paced and auto-weighted strategy to reduce the risk of trapping into bad local optima. Huang et al. [151] applied a soft-weighting scheme of self-paced learning for instances to mitigate the negative impact of noises and outliers. Additionally, they designed a self-paced feature selection manner and a weighting term for views to alleviate the feature and view quality issues.

3.9 Discussion

Although numerous MVC algorithms have been proposed, there is no criterion to decide which MVC algorithm is the best due to the unique merits of each approach. In summary, NMF-based approaches provide straightforward interpretability for clustering results—that is, each observation can be explained as an additive linear combination of non-negative basis vectors. However, a challenge lies in limiting the search of factorizations to those that can give meaningful and comparable clustering solutions across multiple views simultaneously. Graph-based approaches can learn the correlation and complex structures hidden in data, but the performance is highly dependent on some prior factors. Subspace learning is effective in reducing the "curse of dimensionality," but they have initialization dependence. Deep learning based approaches can explore the relationships among different samples and avoid the possible corruption as well as the curse of dimensionality, but many deep learning based MVC approaches involve more parameters and lack theoretic interpretability. Co-clustering can use samples to induce feature clustering and use features to induce sample clustering based on the duality, but co-clustering-based MVC approaches usually suffer from high computational and storage complexities. Self-paced learning approaches avoid bad local minima, but it is difficult to distinguish 'easy' and 'complex' examples.

Several MVC approaches integrated different techniques to enhance the reliability of clustering results via inheriting the merits of different approaches. For example, Tao et al. [152] exploited several candidate MVC to maximize the worst-case performance gain against the best single-view clustering. This ensures that the clustering performance, when utilizing multiple views, is never statistically significantly worse than that achieved by using a single view alone. Yuan et al. [153] designed a self-tuning MVC approach that introduced a sum-of-norm loss function, regularization,

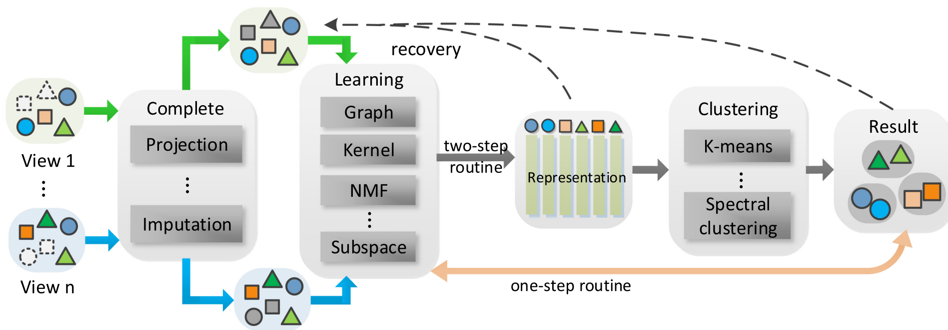

Fig. 7. General procedure of the IMVC approaches.

and statistics techniques to reduce the initialization sensitivity and automatically determine the cluster number.

4 INCOMPLETE MULTI-VIEW CLUSTERING

The problem of clustering incomplete multi-view data is known as IMVC (or partial multi-view clustering) [154]. The goal of IMVC is to discover the common cluster patterns hidden in incomplete multi-view data based on the observed data samples and features in different views. In incomplete data, due to the degradation of the acquired data quality, the natural alignment property of same samples across multiple views and sample completeness may not be preserved [155]. This may lead to a series of information loss, the information imbalance aggravation among different views [19], and the integration difficulties of multiple views [156]. Consequently, it results in the insufficient excavation of the complementary and consistent information [157]. IMVC faces more challenges compared to MVC. It needs issues such as mitigating the impact of missing information and effectively extracting and utilizing the underlying semantic information from missing views/features to cluster the multi-view data. The existing approaches for MVC cannot be directly applied to incomplete multi-view data.

An intuitive strategy to solve IMVC is data adaption—that is, removing incomplete samples or filling incomplete views (imputation) to obtain complete multi-view data. Subsequently, existing MVC methods can be employed directly. The removal of incomplete samples is simple and straightforward, but it loses original samples and may cause heavy information loss when most of the samples have missing values. Imputation methods include zero imputation, mean imputation, k-nearest-neighbor imputation, random imputation, regression imputation, expectation maximum (EM) imputation, multiple imputation (imputes the missing value multiple times, MI), and so on [158]. Because inaccurate interpolation or filling of missing data may induce noisy features, which may destroy the distribution of original data and damage the clustering performance [159], imputation-free approaches have been developed. For example, Chao et al. [160] dealt with the missing samples or features by introducing a sample-view indicator matrix to indicate whether a sample exists in a view or not.

In recent years, a number of approaches for IMVC have been developed. In this section, we present the working mechanisms of the representative IMVC approaches. The general procedure of the approaches for IMVC is shown in Figure 7, and the characteristics of the representative IMVC approaches are summarized in Appendix Table 2.

4.1 NMF-Based Approaches for IMVC

To acquire a robust low-dimensional consensus latent representation for all views, several researchers [161–164] adopted a graph-regularized matrix factorization model to reveal the geometrical structure of data. Specifically, Wen et al. [161] designed a semantic consistency constraint to reduce the influence of the unbalanced incomplete views, Liang et al. [162] and Liang et al. [163] used orthogonal constraint to alleviate the problem of the inconsistent clustering structure, and Liu and Song [164] utilized the guidance of the virtual label to enhance the distinctiveness of learned consensus latent representation. Liu et al. [165] combined NMF with a low-rank tensor to capture the higher-order and complementary information. Zong et al. [25] employed NMF to separately handle individual clusters/instances, and introduced locally geometrical information to reduce the negative impact caused by multi-view interaction. Zhou et al. [166] and Liu et al. [167] incorporated index matrices of missing samples into matrix factorization and introduced graph Laplacian regularization to promote the compactness of the low-dimensional representation. Yin and Sun [168] utilized the results of NMF with Laplacian regularization to reconstruct missing views, and the reconstructed views were also used to seek the latent representation. Chao et al. [158] adopted multiple imputations to deal with missing values under the consideration of the missing value uncertainty and designed a view weighting strategy to ensemble the clustering results from multiple views.

To reduce the computational and memory costs, Hu and Chen [169] employed regularized matrix factorization and weighted matrix factorization, and Yin and Sun [170] combined NMF with the graph Laplacian of the cosine similarity matrix directly computed from the original data space.

4.2 MKL-Based Approaches for IMVC

Liu et al. [171] encouraged incomplete kernel matrices to mutually complete each other and integrated imputation as well as clustering into a unified learning procedure. In another work, Liu et al. [172] imputed each incomplete base matrix and introduced prior knowledge (the consensus clustering matrix is required to lie in the neighborhood of a pre-specified one) to regularize the consensus clustering matrix. Li et al. [173] dynamically generated a consensus proxy under the guidance of a shared cluster matrix to improve the effectiveness of imputation and clustering, and preserved sufficient kernel details via adjusting the size of base partitions for further improving the quality of the consensus proxy. Xia et al. [174] employed a similarity graph to guide the kernel completion for capturing the global structure and local non-linear relationship of samples in kernel space.

4.3 Graph-Based Approaches for IMVC

To alleviate the effect of missing data, Wen et al. [175] designed a locality-preserved reconstruction term to infer the missing views. Yang et al. [176] exploited the intrinsic structure of data to complete the missing views. Liang et al. [162] and Liang et al. [163] employed cross-view feature transformation to complete the unobserved samples and incorporated the completion process into multi-view graph learning. Wen et al. [177] designed the feature space based missing-view inferring to recover the missing views. Zhao et al. [178] designed a complete structure inferring strategy to learn the complete structures of all views for disclosing the real distribution of the absent samples under the reconstruction constraint. In addition, Wang et al. [179] transferred feature missing to similarity missing and then used average similarity values in other views to complete the missing similarity entries based on the spectral perturbation theory. Yin and Jiang [180] employed existing data to reconstruct incomplete data and incorporated the reconstruction objective with self-representation-based spectral clustering. On the contrary, Zhang et al. [181] combined a distance regularization term as well as low-rank representation-based non-negativity constraints to directly learn graphs from raw data without filling. Zheng et al. [182] used the available data of each view to learn a corresponding view-specific partial graph and designed a cross-view graph fusion term to learn a consensus complete graph for different views.

To distinguish the contributions from different views for alleviating influence of improper views to the quality of the fused consensus graph, Lv et al. [183] considered the view's importance to learn the affinity matrix of the view. Liang et al. [184] learned sample-level auto weight to fuse graphs. Wen et al. [185] introduced adaptive weights to balance the importance of different views. Li et al. [186] fused incomplete base partition matrices in an auto-weighted manner to generate a unified partition matrix. Liang et al. [187] adopted an adaptive weighting strategy to alleviate the negative impact of unbalanced incomplete views. In addition, they devised a local and global co-regularization to reveal the mutual effect between local clustering from different incomplete views and global clustering across all views.

To handle data with complex distributions and the non-linear information, Zhao et al. [188] exploited spectral analysis to supervise the common representation extracted from all of the views. Wen et al. [189] investigated the local information within each view as well as the semantic consistent information shared by all views. They introduced a co-regularization constraint to minimize the disagreement between the common representation and the individual representations with respect to different views. Zhuge et al. [190] built a similarity matrix to measure the relationships of present instances, and employed a spectral-based method to learn a common probability label matrix and low-dimensional representations of present samples. Wen et al. [191] employed the graph learning and spectral clustering techniques to learn the common representation. In another work, Wen et al. [192] incorporated the feature space based missing-view inferring and manifold space based similarity graph learning, and imposed a low-rank tensor constraint to capture the high-order correlations of multiple views. Deng et al. [193] employed projection learning to reduce the effect of the information imbalance between different views caused by the diversity dimensions and imposed a graph regularization penalty term to capture the geometric structure of data.

In addition, Yu et al. [194] and Wang et al. [195] designed improved anchor selection strategies to choose representative anchor points. These anchor points were crucial for learning the consensus instance-to-anchor similarity matrices for all views and constructing view-wise complete anchor graphs as well as fused complete anchor graphs. Guo and Ye [196] and Yin et al. [197] utilized anchors to compute the similarities between all data points and executed anchor-based spectral clustering to obtain the clustering result. He et al. [198] proposed a structural anchor-based similarity learning model to obtain the intra-view similarity matrix. They employed the paired anchor samples to compute the inter-view similarities, then designed a complete anchor-inferred graph learning scheme to improve efficiency and performance of the spectral clustering. Ji et al. [199] independently built the similarity matrix of each view via the adaptive neighbor assignment strategy to eliminate the necessary of adjusting parameters. Additionally, they employed a non-iterative approach to reduce the computational complexity.

4.4 Subspace-Based Approaches for IMVC

To alleviate the effect of missing data, Li et al. [200] employed affinity matrices learning and tensor factorization regularization to recover the missing views and the subspace structure. They introduced hypergraph-induced hyper-Laplacian regularization to preserve the high-order geometrical structure of data. Yin et al. [201] employed the available instances within views to infer the missing information and adopted a weighted alignment of projection matrices corresponding to different views to learn a discriminative shared embedding. Han et al. [202] employed a reconstruction term to recover missing samples from non-missing ones and decomposed the self-expressiveness coefficient learned from the recovered complete multi-view into a consistent part together with a specific part. This approach aims to reveal the similarity information of view-paired and obtain the unique information of each view. Zhang et al. [203] recovered the missing data based on shared latent representation and learned multi-level graphs of recovered views by self-representation. Additionally, it introduced a tensor nuclear norm regularizer to pursue the global low-rank property and explore both intra-view and inter-view correlations.

To obtain more accurate subspace representation, Zhao et al. [24] described multi-view data in a local manner to obtain clear block-diagonal structure for data distribution and more accurate subspace representation. Hu et al. [204] devised the soft block-diagonal-induced regularizer to fuse and construct a shared representation for all views. It also inserted multiple indicator matrices into the multi-view self-representation model to achieve clustering results. Li et al. [205] utilized a tensor nuclear norm regularizer to diffuse the information of multi-view block-diagonal structure across different views. Niu et al. [206] utilized low-rank matrix factorization to obtain a consensus representation matrix. It then then combined with the objective function of non-negative embedding and spectral embedding subspace clustering for joint optimization. Yu et al. [207] learned missing instances and self-representations in the latent space for capturing the features of missing instances and preserving the original latent spatial structure of the data. It imposed a completeness constraint to guarantee that the learning direction of missing instances was as close to the original data as possible. Zhon and Pun [208] learned a similarity graph for each view rather than a consensus graph and dealt with the incompleteness of the views by fusing information from different views in partition space. Liu et al. [209] jointly performed data imputation and self-representation learning.

To improve robustness against the incompleteness and noises, Fang et al. [210] designed a view evolution scheme to deal with unbalanced incomplete view data (i.e,. different views often have distinct incompleteness) and devised low-rank representation to recover the data. Zhao et al. [211] developed a weighted subspace learning mechanism with low rank and sparse constraints to capture relationship between the data samples.

4.5 Deep Learning Based Approaches for IMVC

Wei et al. [212] optimized multiple groups of decoder deep networks to obtain the completion of data view and multiple shared representations, so as to generate multiple clustering results with high diversity and quality. Xue et al. [213] utilized multi-view auto-encoder to infer the missing features of incomplete samples, and conducted adaptive graph learning as well as graph convolution to extract data structure. Zhao et al. [156] developed the auto-encoder with the manifold alignment constraint and consistency alignment constraint, aiming to preserve the compact inherent local structure within the view and the consistency semantics between incomplete views. Wang et al. [214] combined discriminator networks with auto-encoders to learn common low-dimensional representations as well as the shared cluster structure across multiple views, and employed adversarial training to generate possible values of missing features. Wen et al. [215] designed view-specific encoders to extract the high-level information of multiple views, and introduced a self-paced strategy and a weighted fusion layer to select the most confident samples and obtain the consensus representation shared by all views. Xu et al. [159] and Xu et al. [216] employed auto-encoders to learn features for each view and utilized an adaptive feature projection to avoid the imputation and fusion for missing data.

Xie et al. [217] utilized convolutional neural networks to extract deep features and employed the attention mechanism to fuse these deep features in a weighted way. Wen et al. [218] used several view-specific graph convolutional encoder networks to reveal the high-level features and high-order geometric structure information of data. Xu et al. [219] integrated element-wise reconstruction and the generative adversarial network (GAN) to infer the missing data, and employed multi-layer non-linear transformations to learn high-level common representation. Wang et al. [220] transferred known similar inter-instance relationships to the missing view and adopted graph networks constructed on the transferred relationship graph to infer missing data. It devised view-specific encoders and an attention-based fusion layer to extract the recovered multi-view data and obtain the common representation. Tang and Liu [221] proposed a bi-level optimization framework to dynamically impute missing views from the learned semantic neighbors and automatically select imputed samples for training.

4.6 Contrastive Learning Based Approaches for IMVC

Lin et al. [222] and Lin et al. [223] employed contrastive learning with the maximal mutual information across different views to obtain consistent representation. They used additional prediction networks with minimal conditional entropy of different views to recover the missing data. Wang et al. [224] designed augmentation-free graph contrastive learning and cross-view graph consistency learning to maximize the mutual information of different views within a cluster based on the graphs transferred from the inter-sample relation graphs.

Yang et al. [155] proposed a unified contrastive learning paradigm to simultaneously solve the partially view-unaligned problem and the partially sample-missing problem, where the available pairs were used as positives and some cross-view samples were randomly selected as negatives, and the influence of the false negatives caused by random sampling was alleviated via noise-robust contrastive loss. Xue et al. [225] developed a multi-view unified and specific encoding network to fuse different views into a unified representation, designed diversified graph contrastive regularization to enhance the discriminating power of the learned representation and reduce the information loss caused by the view missing, and utilized robust contrastive learning loss to reduce the effect of noise and unreliable views.

4.7 Discussion

In IMVC, most approaches usually tend to impute/recover/infer values for the missing data of incomplete multi-view data sets in original data space, and then explore cluster information. The imputation quality depends on the estimation of the distribution of available data. Because incomplete multi-view data usually have biased data distribution, the imputation of missing data might induce noise even incorrect information, especially for data with a large missing rate. Chao et al. [158] and Yin and Sun [168] complete missing information to capture the features of missing instances. This approach retains the original latent spatial structure of the data, but the common features of multiple views learned from the complete data may have large variances in both inter-class and intra-class. Because the clustering result is determined by a whole similarity matrix, the imputation has an impact on the clustering of all samples, no matter whether a sample is complete or not. When an imputation is not of high quality, it could adversely affect the clustering performance of all samples, especially for those with complete views. In addition, the filled values usually lack physical meaning and the imputation of missing data is time consuming.

The NMF-based approaches learn a consensus representation shared by all views for clustering. They explore certain information among the observed views and reduce the negative influence of the missing views via the partial view aligned or weighted regularization strategy. The graph-based approaches transform the feature missing problem into the graph space and explore the similarity information among the observed instances in all views to learn orthogonal consensus representation. These representations are more robust to noise and are suitable to the non-linear separable data.

It is difficult for IMVC approaches adopting shallow models to deal with the dependence, as well as the discrepancy among different views and non-linear relations among samples. This limitation restricts their capability to learn discriminative feature representations. Meanwhile, some of them suffer from high computation costs (e.g., in inverse operations of matrices). Compared with the approaches adopting shallow models, deep learning based approaches have the potential to extract more salient features from the data and thus obtain very promising results.

Many IMVC approaches handle the incomplete multi-view data in a two-stage process—that is, first exploring the consistency among multiple views from the complete part of data, then extending the learned consistency to the incomplete part of data. However, feature learning performed on partial data might cause the distribution discrepancy between the features of complete data and that of incomplete data, resulting in the degradation of model generalization capability.

The unification of recovery of the missing views, the representation learning, and the clustering task provide a novel insight to IMVC [226]. However, these approaches often face challenges associated with high computational and storage complexities, limiting their applicability to large-scale datasets. Several novel approaches have been proposed. For example, Zhang et al. [227] integrated missing view imputation into the fuzzy clustering process to realize cooperative learning between the visible and hidden views. They established hidden links between missing views, hidden views, and complete views to improve the quality of the imputed missing views as well as the learned hidden views. Additionally, an adaptive view weighting mechanism was introduced to improve the robustness of the model. Yang et al. [228] robustly learned the common compact binary codes for incomplete multi-view features and optimized the cluster structures in an online fashion. Fang et al. [229] introduced genetics to clustering algorithms to simultaneously learn the consistent information and the unique information based on the subspace decomposition.

5 UNCERTAIN MVC

Uncertain data clustering must first model uncertain data by using either of fuzzy model, evidence-oriented model, or probabilistic model. Moreover, similarity measure plays an imperative role [230]. When two distributions of two uncertain data are heavily overlapped in locations, the geometric distance-based similarity function cannot correctly capture the change between uncertain data with their distributions. On the other hand, similarity measure based on probability distribution, such as the divergence-based similarity function, cannot discriminate the change between data points when they are not closed to each other or completely separated.

Sharma and Seal [230] combined a self-adaptive mixture similarity function composed of geometric distance and S-divergence with k-medoids MVC to reduce the adverse effect of outliers and noises. Those authors [26] also integrated induced kernel distance and Jeffrey divergence in terms of the degree of overlap concerning each view in a dataset to construct a self-adaptive mixture similarity measure (SAM). Subsequently, they developed a multi-view spectral clustering algorithm with SAM as well as pairwise co-regularization to group uncertain data. Duan [231] developed an approximate Bayes approach to directly estimate the cluster assignment and co-assignment probabilities in the range with several different clustering patterns. Chang et al. [232] proposed a Bayesian probabilistic model via variation inference to automatically learn the multiple expert views, multiple clustering structures, and expert confidences. This approach enables the discovery of various ways to cluster data, considering potentially diverse inputs from multiple uncertain experts. Appendix Table 3 summarizes the characteristics of representative uncertain MVC approaches.

6 DYNAMIC MVC

Most existing MVC approaches fuse all views at one time, but in some applications, the number of views changes with time. For the newly collected views, refusing all views at each time is too expensive to store all historical views [233].

To deal with dynamic change of views, Yin et al. [233] and Zhou et al. [234] developed incremental multi-view spectral clustering (IMSC) to integrate different views one by one in an incremental way, where Zhou et al. [234] first learned an initial model from a small number of views, next updated the model when a new view was available, and then used the updated model to learn a consensus result. However, Yin et al. [233] opted to store a single consensus similarity matrix, representing the structural information of all historical views. Once the newly collected view is available, the consensus similarity matrix is reconstructed by learning from its previous version and the current new view. Yin et al. [233] also incorporated sparse and connected graph learning to reduce the noises and preserve the correct connections within clusters.

In addition, Huang et al. [235] designed an alternative iterative algorithm based on the introduction of an incremental learning mechanism to incrementally update the feature selection matrix. This feature selection was then incorporated into an extended weighted NMF to learn a consensus clustering indicator matrix.

There is a major challenge for incremental clustering called the *stability-plasticity dilemma* [236]. On one hand, the clustering results should be plastic to the new input data from non-stationary distributions. On the other hand, the clustering results should retain the performance of previous input data. Appendix Table 4 summarizes the characteristics of representative dynamic MVC approaches.

7 DATASET

Multi-view datasets widely used by most MVC approaches for evaluating clustering performance consist of five categories: text, image, text-gene, image-text, and video.

7.1 Text Datasets

The text datasets consist of news datasets (3Sources, BBC, BBCSport, Newsgroup), multi-lingual documents datasets (Reuters, Reuters-21578), a citations dataset (Citeseer), a WebKB webpage dataset (Cornell, Texas, Washington, and Wisconsin), an articles dataset (Wikipedia), and a diseases dataset (Derm). The statistics of text databases are reported in Appendix Table 5.

7.2 Image Datasets

The image datasets consist of facial image datasets (Yale, Yale-B, Extended-Yale, VIS/NIR, ORL, Notting-Hill, YouTube Faces), handwritten digits datasets (UCI, Digits, HW2source, Handwritten, MNIST-USPS, MNIST-10000, Noisy MNIST-Rotated MNIST), object image datasets (NUS/WIDE, MSRC, MSRCv1, COIL-20, Caltech101), Microsoft Research Asia Internet Multimedia Dataset 2.0 (MSRA-MM2.0), natural scene datasets (Scene, Scene-15, Out-Scene, Indoor), a plant species dataset (100leaves), animal with attributes (AWA), a multi-temporal remote sensing dataset (Forest), a fashion (e.g., T-shirt, Dress, and Coat) dataset (Fashion-10K), a sports event dataset (Event), and image datasets (ALOI, ImageNet, Corel, Cifar-10, SUN1k, Sun397). The statistics of image databases are reported in Appendix Table 6.

7.3 Text-Gene Dataset

The prokaryotic species dataset (Prok) is a text-gene dataset, which consists of 551 prokaryotic samples belonging to four classes. The species are represented by one textual view and two genomic views. The textual descriptions are summarized into a document-term matrix that records the TF-IDF reweighted word frequencies. The genomic views are the proteome composition and the gene repertoire. The statistics of text-gene databases are reported in Appendix Table 7.

7.4 Image-Text Datasets

The image-text datasets consist of Wikipedia's featured articles dataset (Wikipedia), the drosophila embryos dataset (BDGP), the NBA-NASCAR Sport dataset (NNSpt), indoor scenes (SentencesNYU v2 (RGB-D)), the Pascal dataset (VOC), an object dataset (NUS-WIDE-C5), and photographic images (MIR Flickr 1M). The statistics of image-text databases are reported in Appendix Table 8.

7.5 Video Datasets

The video datasets consist of the actions of passenger dataset (DTHC), pedestrian video shot dataset (Lab), motion of body sequences (CMU Mobo) dataset, face video sequences datasets (YouTubeFace_sel, Honda/UCSD), and Columbia Consumer Video dataset (CCV). The statistics of video databases are reported in Appendix Table 9.

8 PERFORMANCE METRICS

The procedure for evaluating clustering results is known as cluster validity. Well-known performance metrics for evaluating the performance of MVC include internal validation indexes and external validation indexes [230].

8.1 Internal Indices

Internal indices are quality scores computed by employing only the information inherent to the dataset, such as Compactness (the average distance between every pair of data points), the Davies-Bouldin Index (the ratio of the sum of within-cluster scatters to between-cluster separations), the Dunn Validity Index (inter-cluster distances over intra-cluster distances), or Separation (the mean Euclidean distance between cluster centroids) [237]. Ideally, the data samples of each cluster are to be as close as possible. Thus, a smaller value of Compactness indicates more compact and better clusters. Separation quantifies the magnitude of separation between the clusters. A lower value of Separation represents the closeness of clusters. Further, the Davies-Bouldin Index identifies the overlapping of clusters by calculating the fraction of the sum of within-cluster distribution to between-cluster separations. A value near to 0 of the Davies-Bouldin Index denotes that the resultant clusters are well separated and compact. Last, the Dunn Validity Index measures the ratio of inter-cluster distance to intra-cluster distance. A higher value of the Dunn Validity Index illustrates well-separated and compact clusters [230].

8.2 External Indexes

The external indexes are quality scores computed by comparing clustering labels with the external supplied true labels, such as **Clustering Accuracy (ACC), Normalized Mutual Information (NMI)**, Purity, the Rand Index (RI)/Adjusted Rand Index (ARI), Precision, Recall, and F-score (F1) [51, 238]. For all external evaluation metrics, a higher value close to 1 of each external index indicates preferable clustering results, whereas a lower value close to 0 denotes undesirable clustering results.

The formulas of evaluation indicators are shown in Appendix Table 10.

9 EMPIRICAL EVALUATION

In this section, we empirically evaluate the performance of 35 approaches, including 29 complete MVC approaches and 6 IMVC approaches. The 29 complete MVC approaches consist of two single-view clustering approaches (SCBest and SCCat). SCBest involves conducting spectral clustering on each single view independently, and the result of the view with the best clustering performance is reported. SCCat involves concatenating vectors from different views into a new vector and then applying the spectral clustering algorithm straightforwardly on the concatenated vector. Additionally, there are 6 NMF-based approaches (MultiNMF [16], MultiGNMF [239], MVCC [44], MVCF[240], MvCDMF [63], and MCDCF [58]), 5 graph-based approaches (SwMC [30], AMGL[32], MCGC [241], RMSC [242], and AWP [243]), 9 subspace-based approaches (DiMSC [244], ECMSC

[245], FMR [107], SM2VC [95], CSMSC [246], DSS-MSC [97], DMSCN [247], MvDSCN [109], and MvSC-MRAR [114]), 3 deep learning based approaches (SDMVC [248], CoMVC [127], and MVC-MAE [117]), 1 contrastive learning based approach (NMvC-GCN [129]), and 3 co-learning-based approaches (CoregSC [15], TW-Co-*k*-means [249], DWMVC [37]). The 6 IMVC approaches consist of 1 graph-based approach (PIMVC [193]), 1 subspace-based approach (LATER [203]), 3 deep learning based approaches (DIMVC [159], APADC [216], DSIMVC [221]), and 1 contrastive learning based approach (COMPLETER [222]). These approaches are representative MVC approaches, covering the main categories in the taxonomy proposed in this article and having different structures or constraints.

The experiments are conducted on seven datasets, including three text datasets (BBCSport, Reuters, Reuters-21578), three image datasets (NUSWIDE, Caltech101-7, NUSWIDE30K), and one image-text dataset (RGB-D). NUSWIDE30K contains 30,000 instances, whereas the other datasets contain hundreds or thousands of samples. These datasets are widely recognized as well-known benchmarks in MVC communities, representing different typical application scenarios, each with distinct views, classes, instances, features, and feature dimensions. We choose them to explore how various approaches behave in different types of datasets. ACC and NMI are used as performance metrics. In the experiments, all algorithms are run five times on each dataset and the average performance are reported. All evaluations are carried out on a standard Ubuntu-18.04 OS, and 11 approaches (DMSCN, MvDSCN, MvSC-MRAR, SDMVC, CoMVC, MVC-MAE, NMvC-GCN, COMPLETER, DSIMVC, DIMVC, APADC) are implemented by using Pytorch 1.0 with an NVIDIA 2080Ti GPU, whereas other approaches are implemented by using Matlab with an Intel Core i7-7820X CPU. For each compared approach, a grid search is performed in the parameter set suggested in the corresponding work and the best results are reported.

The ACCs and NMIs of 29 complete MVC approaches and 6 IMVC approaches are shown in Appendix Table 11 (complete MVC) and Appendix Table 12 (IMVC), where bold numbers represent the best results in the column. From Appendix Table 11 and Appendix Table 12, we have the following observations and analyses:

(1) The clustering performance of SCBest and SCCat are not necessarily worse than that of MVC algorithms, especially SCCat achieves the second-highest ACC and NMI values on the BBCSport dataset. Such phenomena are contrary to the general expectation that multi-view algorithms can obtain better performance than that of merely using a single view. This indicates that it is important to design a strategy for integrating multi-view information; otherwise, a simple concatenation may perform better.

(2) Among all MVC approaches, MvSC-MRAR achieves the highest ACC and NMI values on Reuters, Reuters-21578, NUSWIDE, Caltech101-7, and RGB-D; meanwhile, it also achieves the third-best clustering performance, after that of NMvC-GCN and SCCat, on the dataset BBCSport. Among the nine subspace-based MVC approaches, approaches with deep structure generally outperform those with shallow structure. However, MvDSCN and DMSCN do not perform as well as MvSC-MRAR, because they only use the information of the last layer in encoder components, without integration of information from different levels. This indicates that deep learning and the integration of the non-linear structure, multi-level representation information in multi-view data, and data distribution of latent representation are helpful for improving the performance of MVC. In addition, NMvC-GCN achieves the highest ACC and NMI values on BBCSport, and its ACC and NMI values are also close to those of MvSC-MRAR on other datasets. This indicates that contrastive learning is effective in MVC.

(3) No one MVC approach can maintain consistent good performance on various datasets. For example, MultiGNMF with local graph regularization achieves the highest ACC and NMI

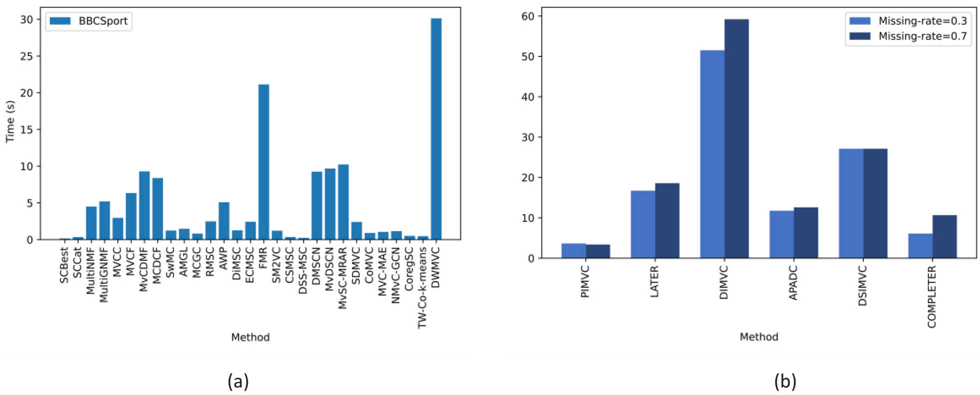

Fig. 8. Comparison of the runtime of different algorithms on the BBCSport dataset (complete/incomplete).

values on Caltech101-7 and RGB-D, whereas MultiNMF with the consensus constraint and MVCC with multi-manifold regularization achieve the highest ACC and NMI values on BBCSport and Reuters, respectively. These phenomena show that factors such as model structures, regularization constraints, and weights corresponding to different views all contribute to improving clustering performance. Moreover, the information captured for clustering under different conditions is different.

(4) Except for the single-view approaches (SCBest, SCCat), the deep learning based approaches (SDMVC, CoMVC, MVC-MAE), AWP, and TW-Co-*k*-means, other approaches did not complete the clustering task (i.e., have not generated clustering results) on the dataset of NUSWIDE30K, due to running out of memory or heavy time consumption. However, the ACC and NMI values obtained by SCBest, SCCat, SDMVC, CoMVC, MVC-MAE, AWP, and TW-Co-*k*-means on NUSWIDE30K are relatively low. This suggests that these approaches do not fully capture the inherent geometrical structure of NUSWIDE30K. The challenge of clustering large-scale multi-view data persists.

(5) In Appendix Table 12 (IMVC), LATER achieves the highest ACC and NMI values on BBCSport, Reuters-21578, and RGB-D under various missing-rates (ratio of missing samples to all samples), but an exception occurs on Reuters, and DSIMVC achieves the highest ACC and NMI values on NUSWIDE and NUSWIDE30K under various missing-rates. On Caltech101-7, COMPLETER outperforms other approaches when the missing-rate is 0.1, whereas APADC does well when the missing-rate is greater than 0.5, and the performance of DIMVC and PIMVC is not as good as other approaches. Furthermore, we also see that ACC and NMI values of many algorithms decrease with increase of the missing-rate.

Figure 8 shows the runtimes of different approaches on the BBCSport dataset. From Figure 8(a) (complete MVC), it can be seen that SCBest, SCCat, CSMSC, DSS-MSC, TW-Co-*k*-means, and CoregSC run well, and SDMVC, CoMVC, MVC-MAE, NMvC-GCN, SwMC, AMGL, MCGC, RMSC, DiMSC, ECMSC, and SM2VC also run reasonably well. However, MultiNMF, MultiGNMF, MVCC, MVCF, MvCDMF, MCDCF, AWP, DMSCN, MvDSCN, and MvSC-MRAR run slower—especially FMR and DWMVC run unreasonably. The difference in running time between different approaches stems from many factors, such as the design of the algorithms, the complexity of adopted techniques, the clustering routine, the model structure, the number of samples, and the dimension of features, among others. For example, MvCDMF requires multiple layers of deep NMF decomposition, and MvDSCN requires multiplication with a matrix of the square of the similarity matrix during each iteration, which leads to longer execution time. In Figure 8(b) (IMVC), DIMVC is the slowest and PIMVC is the fastest. Furthermore, the same approach takes longer on datasets with higher missing-rates.

10 FUTURE WORK

Although existing MVC approaches have obtained promising performance via different strategies, such as the weighting views or samples, imposing manifold or low-rank constraints, there are still some unresolved problems in the field of MVC, which are worthy of the attention of researchers. In the following, we summarize several directions of future research to encourage more research in MVC.

MVC for Large-Scale and High-Dimensional Data. The datasets used for current experimental evaluation are often million-scale with low dimensionality, and only a few of them are billion-scale or have thousands of feature dimensions. However, in many real applications, the datasets may involve billions of samples or have high dimensionality—for example, several million posts are shared per minute on Facebook, and each person has millions of genetic variants as genetic features in bioinformatics [11]. As a result, existing MVC solutions may fail to process such real large-scale and/or high-dimensional data within reasonable time cost. Furthermore, the performance of algorithms is also affected in a high-dimensional situation, because high-dimensional data often has a large amount of redundant information. The redundant information not only fails to supplement valid information but also jeopardizes good data feature representation [6]. It is important to develop approaches to efficiently perform clustering on these large-scale or high-dimensional multi-view data. One possible research direction is developing clustering algorithms based on distributed computation platforms, or keeping the data on disk and designing I/O-efficient clustering algorithms.

MVC for Dynamic and Uncertain Data. Existing MVC mainly focuses on clustering the static data, and the settings of dynamic data clustering are overlooked. In real-life scenarios, data are not always static (e.g., video data). On one hand, some samples appear while other samples disappear. On the other hand, samples may be described by some time-varying information. The techniques for dynamic clustering need to be scalable and better to be incremental so as to deal with dynamic changes efficiently. How to design effective MVC approaches in dynamic domains remains an open question. Moreover, how to integrate various features for uncertain data clustering is also a subject deserving of further study.

MVC for Data with an Unknown Number of Clusters. Most existing MVC works assume that the number of clusters is known. This assumption, however, is too strong in real applications, especially in multi-view scenarios. How to design MVC approaches that can automatically determine the number of clusters is a problem worthy of careful investigation.

MVC with More Interpretation. Most deep learning approaches can explore the relationships among different samples and avoid possible corruption as well as the curse of dimensionality, but many deep MVC approaches involve more parameters and lack theoretic interpretability. Thus, we may need to construct deep networks based on optimization approaches and make the networks more interpretable.

11 CONCLUSION

As a powerful learning tool for exploring the structure of multi-view data, MVC is widely used in many disciplines and plays a crucial role in various applications. However, the complex distribution and diversified heterogeneous features of multi-view data impose challenges for MVC. Despite that several survey studies to assess MVC approaches have been performed in the past, these survey studies mainly focus on limited approaches for complete multi-view data or those for incomplete multi-view data. They often overlook approaches for complete, incomplete, uncertain, and dynamic multi-view data at the same time. To fill this gap and provide an updated survey for MVC approaches including those proposed after 2019, this study first analyzed and introduced the basic concepts and common key techniques for MVC, then proposed a novel taxonomy of MVC approaches. The study then presented the working mechanisms as well as characteristics of the representative MVC approaches proposed in recent years. Following this, the article summarized representative MVC datasets and performance metrics commonly used in the MVC field. Finally, the study selected 35 representative MVC approaches from the proposed taxonomy to conduct an empirical evaluation on seven real-world benchmark datasets.

The study provided theoretical knowledge of MVC, a novel taxonomy for MVC approaches, and insightful information on performance features of existing approaches. Its primary goal was to assist researchers interested in MVC and its related areas by offering an in-depth understanding on the current progresses of MVC. Additionally, it aimed to serve as an insightful guideline to practitioners for application development.

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